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Groin problems from pre- to in-season: a prospective study on 386 male Spanish footballers

Ernest Esteve[a,b], Michael Skovdal Rathleff[c,d], Per Hölmich[e], Martí Casals[f,g], Mikkel Bek Clausen[e,h], Jordi Vicens-Bordas[b,f], Tania Pizzari[i] and Kristian Thorborg[e,j]

[a]School of Health and Sport Sciences (EUSES), Universitat de Girona, Salt, Spain; [b]Sportclínic, Physiotherapy and Sports Training Centre, Girona, Spain; [c]SMI, Department of Health Science and Technology, Faculty of Medicine, Aalborg University, Aalborg, Denmark; [d]Department of Occupational Therapy and Physiotherapy, Aalborg University Hospital, Aalborg, Denmark; [e]Sports Orthopedic Research Center (SORC-C), Department of Orthopedic Surgery, Copenhagen University Hospital, Copenhagen, Denmark; [f]Sport and Physical Activity Studies Centre (CEEAF), Faculty of Medicine, University of Vic – Central University of Catalonia, Barcelona, Spain; [g]Medical Department, Futbol Club Barcelona, Barça Innovation Hub, Barcelona, Spain; [h]Department of Midwifery, Physiotherapy, Occupational Therapy and Psychomotor Therapy, Faculty of Health, University College Copenhagen, Copenhagen, Denmark; [i]Department of Rehabilitation, Nutrition and Sport, La Trobe University Sport and Exercise Medicine Research Centre, Bundoora, Australia; [j]Physical Medicine and Rehabilitation – Copenhagen (PMR-C), Amager-Hvidovre Hospital, Copenhagen University, Hvidovre, Denmark

ABSTRACT

This study investigated the weekly prevalence of groin problems over a 3-week football pre-season, compared to a 39-week competitive in-season. We registered time-loss groin injuries, and self-reported weekly groin-pain in 17 amateur male football teams (386 players). The average weekly prevalence of groin problems (prevalence ratio (PR)) was 1.8 times higher (95% CI 1.6 to 2.0) during pre-season (21%) compared to in-season (12%). We found a higher weekly prevalence (PR 1.8; 95% CI 1.6 to 2.1) of groin problems without time loss, during the pre-season (19%) compared to the in-season (10%), but no significant difference in the weekly prevalence of groin problems with time loss (PR 1.5; 95% CI 1.0 to 2.4). Attention should be given to optimal load progression, and early implementation of preventive measures during the football pre-season to reduce the prevalence of groin problems in both pre- and in-season.

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KEYWORDS
Groin pain; groin injuries; injury prevention; injury epidemiology; HAGOS

Introduction

In men's football, the highest weekly prevalence of groin problems is at the beginning of the in-season (20%) (Esteve et al., 2020), and during periods of increased match frequency (29%) (Harøy et al., 2017). Rapid changes in football loads have been shown to increase the risk of injury in professional football players (Malone et al., 2017), with low levels of off-season sport-specific training increasing the risk of a new groin injury when training loads resume (Emery & Meeuwisse, 2001).

CONTACT Ernest Esteve eesteve@euses.cat Sportclínic, Physiotherapy and Sports Training Centre , Girona 17003, Spain

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Few prospective epidemiological studies investigate the prevalence of groin problems using groin symptoms and self-reported functional impairment in addition to time loss (Esteve et al., 2020; Harøy et al., 2017), with none investigating pre-season. The present study compared the extent of groin problems between the pre- and the in-season in amateur Spanish male players. We also investigated if limitations in hip- and groin-related sporting function due to groin problems were different between pre- and in-season.

Methods

The in-season data and methodological details for the present prospective cohort study have been published previously (Esteve et al., 2020). Over a 3-week pre-season period, a convenience sample of amateur Spanish players from 17 teams (tiers IV to VI) completed a weekly survey asking about groin pain, irrespective of time loss, during the preceding week. Players also completed the Sport/Recreation subscale from the Hip And Groin Outcome Score (HAGOS) questionnaire measuring hip and groin sporting function (Thorborg, Hölmich et al., 2011) each week. Football staff registered the number and characteristics of time-loss groin injuries and team exposure as minutes of participation in physical activity under the supervision of the coaching staff.

Groin problems with and without time loss were determined using information from time-loss injury records and groin-pain surveys. A player was considered as having a groin problem with time loss when, in a given week, he missed at least one training session or was not available for match selection due to groin injury (i.e., time-loss groin injury). A player was considered as having a groin problem without time loss, when he answered "yes" to the question "Did you suffer from groin pain (inguinal region including adductor, lower abdomen, and pubic region) in the past week?", but no time loss due to a groin problem was reported in that week.

Individual HAGOS, Sport/Rec subscale scores were calculated per player at every time point (3 pre-season time points) regardless of whether players reported groin pain or not in the preceding week.

Ethics approval was obtained from a regional committee. All participants provided written informed consent according to the Helsinki Declaration prior to participating.

Data analyses

Weekly prevalence of groin problems was calculated by dividing the number of players reporting groin problems with or without time loss, respectively, by the number of players who were part of a team in each week (Esteve et al., 2020). We calculated injury incidence (Hägglund et al., 2009), and injury burden of time-loss groin injuries (Ekstrand et al., 2016) using team exposure to football in hours.

We calculated a prevalence ratio (PR) and prevalence difference (PD) to compare pre- with in-season prevalence; and incidence rate ratio (IRR) and rate difference (RD) to compare injury incidence and burden. For comparisons of HAGOS-Sport/Rec scores from players reporting groin problems between pre-season and in-season, we performed a linear mixed model. Season period (pre-/in-season) was included as fixed effect, while player was considered as a random effect.

Data were analysed in IBM SPSS statistics 25.0, and R version 3.3.2 (R Core Team, 2014). Alpha level was set at 5%.

Results

On average, 74.3% of the players responded to the weekly groin-pain survey (Week 1: 82% (n = 309); week 2: 68% (n = 372); week 3: 72% (n = 379)). Figure 1 shows weekly prevalence of groin problems during pre- and in-season. The average weekly prevalence of all groin problems and groin problems without time loss was higher during the pre-season compared to the in-season. For groin problems with time loss, there was no significant difference between pre- and in-season. Neither time-loss groin-injury incidence or injury burden were different in the pre- compared to the in-season (Table 1).

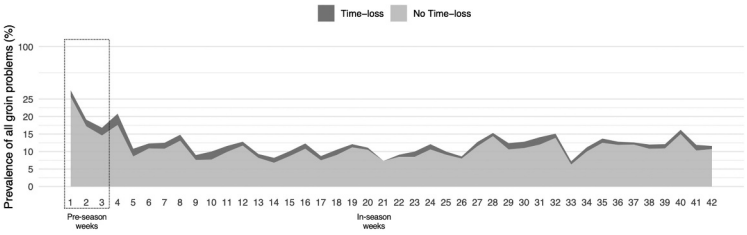

Figure 1. Weekly prevalence proportions of groin problems during pre- and in-season periods.

Table 1. Pre- and in-season data overview and measures compared.

	Pre-season (n = 386)	In-season (n = 407)	Ratios†	Difference‡
Data overview (SD)				
Follow-up	3 weeks	39 weeks		
Total number of participants	386	407		
Age, years	23 (4)	23 (4)		
Weight, Kg	74 (8)	74 (7)		
Height, cm	178 (6)	178 (6)		
Number of players reporting a groin-pain episode[a]	125	216		
Number of time-loss groin injuries	10	63		
Days lost due time-loss groin injury	72	617		
Total exposure, hours	8576	71908		
Average exposure team/week	6.9 h (2)	6 h (1)		
Measures compared (95% CI)				
Weekly prevalence of groin problems	21% (18% to 23%)	12% (11% to 12%)	1.8 (1.6 to 2.0)	9.0% (6.4% to 11.6%)
Groin problems with time loss	2.0% (1.3% to 3.1%)	1.3% (1.1% to 1.5%)	1.5 (1.0 to 2.4)	0.7% (−0.2% to 1.6%)
Groin problems without time loss	19% (16% to 21%)	10% (10% to 11%)	1.8 (1.6 to 2.1)	8.3% (5.9% to 10.8%)
Time-loss groin-injury incidence[b]	1.2 (0.6 to 2.2)	0.9 (0.8 to 1.0)	1.3 (0.7 to 2.6)	0.03 (−0.05 to 0.10)
Time-loss groin-injury burden[c]	8.4 (6.6 to 10.6)	8.6 (7.9 to 9.3)	1.0 (0.8 to 1.3)	0.01 (−0.21 to 0.19)

†Prevalence ratios for prevalence measures, and incidence rate ratios for incidence and burden measures; ‡prevalence difference for prevalence measures and rate difference for incidence and burden measures (pre-season group as reference). [a]Number of players reporting at least one episode of groin pain; [b]number of injuries per 1000 hours of football; [c]number of days lost per 1000 hours of football.

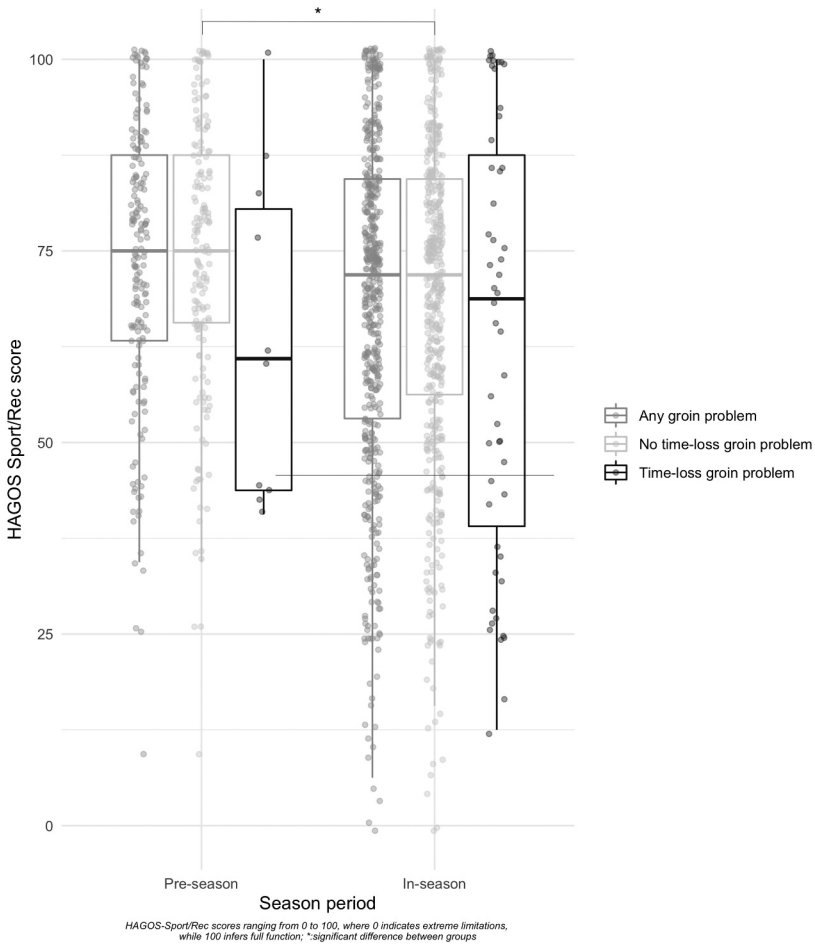

Figure 2. HAGOS – Sport/Rec subscale scores from player reporting any groin problem, and groin problems with and without time loss, from pre- and in-season periods.

In pre-season compared to in-season, there were minor mean differences in HAGOS-Sport/Rec scores among players reporting "any groin problem" (mean difference: 2.7; 95% CI −0.2 to 5.6; P =.06), players reporting time-loss groin problems (mean difference: −2.3; 95% CI −21.0 to 16.4; P = .80); and players reporting groin problems without time loss (mean difference: 3.1; 95% CI 0.3 to 6.0; P = .03), of which only the latter was statistically significant (Figure 2).

Discussion

We found a high (21%) pre-season weekly prevalence of groin problems. Previous studies show similarly high pre-season point-prevalence of groin symptoms (20% to 35%) (Harøy et al., 2019; Langhout et al., 2019; Thorborg, Rathleff et al., 2017). This highlights the need for a greater focus on prevention strategies during pre-season considering the recurrent and persistent nature of groin problems (Thorborg, Rathleff et al., 2017). Asking two

simple questions at pre-season about past-season groin pain and its duration is sufficient to detect players at higher risk of experiencing severe reductions in their hip and groin function at the beginning of the new season (Esteve et al., 2018; Thorborg, Rathleff et al., 2017).

Although the substantial reduction in weekly prevalence of groin problems from pre- to in-season occurred without the initiation of any controlled prevention strategies, the implementation of a hip adductor strengthening exercise during pre-season could further reduce the prevalence of groin problem in the in-season (Harøy et al., 2019). This reflects the importance of pre-season training for developing sport-specific adaptations to protect against in-season groin problems, in addition to improving player performance.

The highest weekly prevalence of groin problems (28%) was registered in the first week of pre-season. This could indicate that prevalence of groin problems may peak as football pre-season starts, emphasizing the importance of balancing and progressing training loads from the off-season to the pre-season (Malone et al., 2017). The off-season period could also provide an opportunity to implement novel measures outside team-training routines aimed at building hip adductor muscle capacity and resolving existing groin problems (Thorborg, Rathleff et al., 2017).

We found minimal mean differences (3 points) on HAGOS-Sport/Rec scores between pre- and in-season and their clinical relevance is debatable. However, worse pre-season hip and groin health has been linked to an increased risk of groin injury in the subsequent in-season in male football players (Bourne et al., 2020). Early detection and close monitoring is advisable to facilitate optimal management and to prevent recurrence and/or aggravation. This could be achieved using specific alerts on hip adductor strength test (>15% reduction), HAGOS questionnaire (score <70 out of 100) (Wollin et al., 2018). The "five-second squeeze test" is an excellent surrogate measure of HAGOS and could be also useful to reflect hip and groin health trough the full season (Thorborg, Branci et al., 2017).

Exposure could only be registered successfully at the team level, leading to slightly underestimated injury rates (Kristenson et al., 2016). Response rate to the groin pain survey (75%) was lower compared to our previously reported in-season rate (93%), but our method of prevalence calculation negates any overestimation. We captured 3 weeks of pre-season, despite varying pre-season lengths: 12 teams had a 4-week pre-season, 2 teams had a 5-week pre-season and 2 teams were included in their last pre-season week. We consider present estimates representative of most European amateur male football levels, although generalizations to extended pre-season and/or higher football levels must be made with caution.

Conclusion

Groin problems were twice as prevalent during pre-season compared to the in-season. This difference was due to a greater weekly prevalence of groin problems without time loss during the pre-season. This highlights the importance of optimal progression of pre-season loading and early implementation of preventive measures to reduce the prevalence of groin problems in both pre- and in-season.

Disclosure statement

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ORCID

Ernest Esteve http://orcid.org/0000-0002-6798-2294
Michael Skovdal Rathleff http://orcid.org/0000-0003-1173-0335
Per Hölmich http://orcid.org/0000-0003-2098-0272
Martí Casals http://orcid.org/0000-0002-1775-8331
Mikkel Bek Clausen http://orcid.org/0000-0003-4125-8975
Jordi Vicens-Bordas http://orcid.org/0000-0002-8388-8863
Tania Pizzari http://orcid.org/0000-0002-8804-0095
Kristian Thorborg http://orcid.org/0000-0001-9102-4515

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